

$$1.1. \quad a) \operatorname{Re}\langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$b) \operatorname{Im}\langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$* \langle u, v \rangle = \operatorname{Re}\langle u, v \rangle + (\operatorname{Im}\langle u, v \rangle) i$$

$$\frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) = \frac{1}{4} (\langle u+v, u+v \rangle - \langle u-v, u-v \rangle) =$$

$$\stackrel{SL2}{=} \frac{1}{4} (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle) =$$

$$= \frac{1}{4} (2\langle u, v \rangle + 2\langle v, u \rangle) \stackrel{SSS}{=} \frac{1}{2} (\langle u, v \rangle + \overline{\langle u, v \rangle}) =$$

$$\stackrel{*}{=} \frac{1}{2} (\operatorname{Re}\langle u, v \rangle + (\operatorname{Im}\langle u, v \rangle) i + \operatorname{Re}\langle u, v \rangle - (\operatorname{Im}\langle u, v \rangle) i) =$$

$$= \frac{2}{2} \operatorname{Re}\langle u, v \rangle \quad \text{Reálna časť sedí, teda už}$$

jen Imaginárna časť

$$\frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2) = \frac{1}{4} (\langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle)$$

$$\stackrel{SL2}{=} \frac{1}{4} (\langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + \overbrace{(-i)(-i)}^1 \langle v, v \rangle - \langle u, u \rangle - i\langle u, v \rangle +$$

$$+ i\langle v, u \rangle - \overbrace{(i \cdot i)}^1 \langle v, v \rangle) = \frac{1}{4} (2i\langle v, u \rangle - 2i\langle u, v \rangle) \stackrel{SSS}{=}$$

$$= \frac{1}{2} (2i\overline{\langle u, v \rangle} - 2i\langle u, v \rangle) =$$

$$\stackrel{*}{=} (\operatorname{Re}\langle u, v \rangle) i - (\operatorname{Im}\langle u, v \rangle) i^2 - (\operatorname{Re}\langle u, v \rangle) i - (\operatorname{Im}\langle u, v \rangle) i^2 = \operatorname{Im}\langle u, v \rangle$$